CONCEPT CFD · FRONT WING IN GROUND EFFECT · 250 KM/H

F1 FRONT WING 2026 — AERODYNAMIC ASSESSMENT

First virtual wind-tunnel run of the three-element 2026 front wing concept (wing, endplates, footplates and nose) running 75 mm above a moving road. The goal is a quick read on how much downforce the concept makes, what it costs in drag, and where the load comes from.

Explore the result in 3-D: interactive CFD viewer (rotate the wing, switch surface pressure and skin friction, show the vortices and the road footprint).

DOWNFORCE @ 250 KM/H	DRAG @ 250 KM/H	EFFICIENCY (DOWNFORCE ÷ DRAG)	-CLA / CDA
1,929 N ≈ 197 kg	**213** N ≈ 22 kg	**9.0**	**0.67** / 0.07 m²

SUMMARY FOR MANAGEMENT

- **What we did.** We simulated the airflow around the front wing concept at 250 km/h with the road moving underneath, as on track. Mesh and solve took about 41 minutes on one workstation.
- **Headline.** The wing produces about **1,929 N of downforce** (the weight of 197 kg pressing the front axle down) for **213 N of drag**. That is 9.0 N of downforce for every newton of drag.
- **Where the load comes from.** The load peaks about 212 mm from the car centreline. The inboard half of each wing (0–450 mm) carries 66 % of the downforce and the outboard half, towards the endplate, 34 %. Most of it comes from suction on the underside of the mainplane, which runs closest to the road.
- **Drag make-up.** 85 % of the drag comes from pressure (shape and vortices) and 15 % from skin friction. Shape changes, not surface finish, are the lever for drag.
- **Confidence.** This is a concept-level run on a deliberately coarse mesh to give an answer within the hour. The downforce still swings by about ±20 % from iteration to iteration (the number quoted is the average), and mesh resolution adds further uncertainty, so treat absolute values as indicative (roughly ±20–25 %). Comparisons between design variants run with identical settings are more reliable than the absolute values.
- **Recommended next step.** A ride-height sweep (front wing height is the main sensitivity in ground effect) and a run with front wheels added, since the wheel–wing interaction drives real-car front wing design.

KEY AERODYNAMIC FINDINGS

1. **Strong ground effect under the mainplane.** Air under the mainplane reaches about 100 m/s, 1.5 times the car speed. This high-speed, low-pressure channel between wing and road is the main source of downforce, and makes the wing sensitive to ride height.
2. **Inboard flow separation (main improvement opportunity).** At 450 mm from the centreline the flow leaves the mainplane underside before its trailing edge, leaving a slow, separated region beneath elements 2 and 3 (dark blue in the 450 mm side cut). That region carries no load and adds drag. Further outboard (600 mm) the flow stays attached through all three elements. Candidate fixes: less incidence or camber on the inboard mainplane, or a larger slot gap / overlap to element 2 so more high-energy air re-attaches the flow.
3. **Outboard elements work as intended.** The 600 mm cut shows clean slot jets and attached flaps, with the wake turned upward, as a multi-element front wing should.
4. **Endplate vortex system is established.** Vortices from the endplate top and footplate edges roll up and travel rearward (see the vortex and streamline images). Their path relative to the front tyre is the next thing to assess once wheels are modelled.

Coarse meshes with wall functions can over-predict separation. Confirm finding 2 on a finer mesh before changing the geometry.

DOWNFORCE AND DRAG VERSUS SPEED

CAR SPEED	DOWNFORCE [N]	DOWNFORCE [KG]	DRAG [N]
100 km/h	309	31	34
150 km/h	694	71	77
200 km/h	1,235	126	136
250 km/h	1,929	197	213
300 km/h	2,778	283	307
330 km/h	3,361	343	372

Forces grow with the square of speed, so doubling speed gives four times the load. This assumes the same ride height at every speed.

READING THIS REPORT

TERM	MEANING
Downforce	Vertical aerodynamic load pushing the car onto the road (gives grip in corners).
−ClA / CdA	Downforce and drag per unit dynamic pressure, in m^2. The usual F1 way to compare aero packages independent of speed.
Efficiency	Downforce divided by drag. Higher means more grip for less straight-line speed penalty.
Cp	Pressure coefficient. Negative (blue) is suction; on the underside it pulls the wing towards the road.
Vortex	Swirling tube of air shed from edges such as endplates. F1 teams use them to steer air around the front wheels.

ENGINEERING DETAIL

OPERATING POINT AND SET-UP

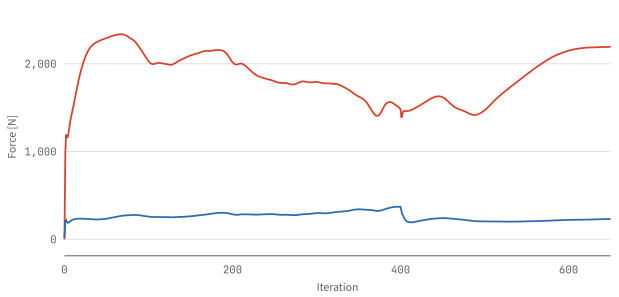
ITEM	VALUE
Geometry	`Front_wing_mesh.stl`: 3-element wing, endplates, footplates, nose (893,672 triangles, mm → m)
Car speed / freestream	69.44 m/s (250 km/h), air $\rho = 1.2$ kg/m^3, $\nu = 1.5\times10^{-5}$ m^2/s
Reynolds number	0.93 M on the 200 mm mainplane root chord
Ground	Road plane at z = 0 (FIA reference plane), moving at car speed. Lowest wing point 75 mm above it
Domain	9.6 × 3.0 × 2.4 m half domain (x −4.8…4.8 m), symmetry at the car centreline, slip top and side walls
Nose closure	Nose base at x = −0.1 m continued as a constant-section chassis stub to x = 1.6 m. Stub is excluded from forces
Turbulence	k-ω SST, wall functions; inlet intensity 1 %, $\nu_t/\nu = 10$
Solver	`foamRun -solver incompressibleFluid` (SIMPLEC), 2nd-order bounded upwind for U, 16 MPI ranks
Stabilisation	About 100 poor cells at the nose / mainplane junction went unstable in the first run. Restarted at iteration 400 with a velocity limiter ($\lvert U\rvert \le 150$ m/s), 1 non-orthogonal corrector and first-order k-ω convection
Reference values	$A_{ref} = 1$ m^2 (coefficients read as ClA / CdA), q = 2,893 Pa, moments about x = −0.6 m

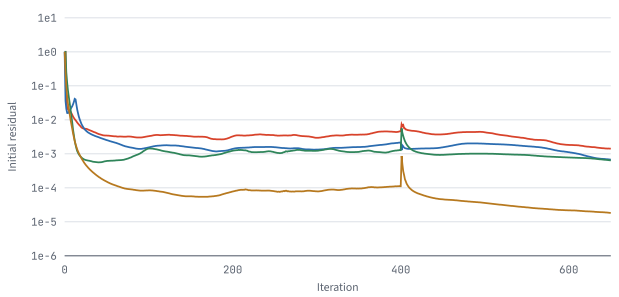
INTEGRATED FORCES (FULL WING)

COMPONENT	DOWNFORCE [N]	DRAG [N]
Pressure	1,931.2	180.8
Viscous (skin friction)	-2.1	32.5
Total	**1,929.1**	**213.2**
Iteration-to-iteration swing, last 162 iterations	±387.7 (±20 %)	±15.3 (±7 %)

Values are averaged over the last 162 iterations and doubled from the half model. Downforce scales with speed squared: at 300 km/h expect about 2,778 N, at 200 km/h about 1,235 N.

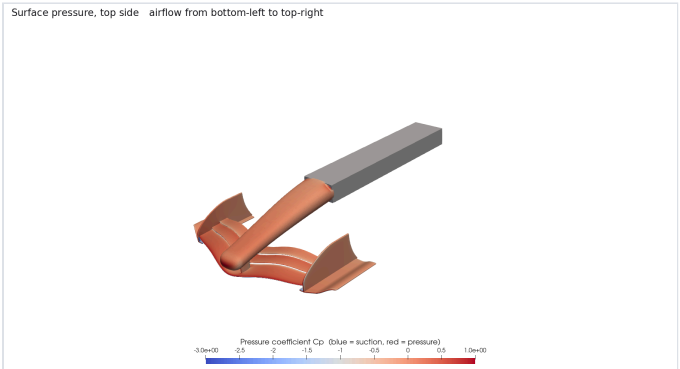
CONVERGENCE



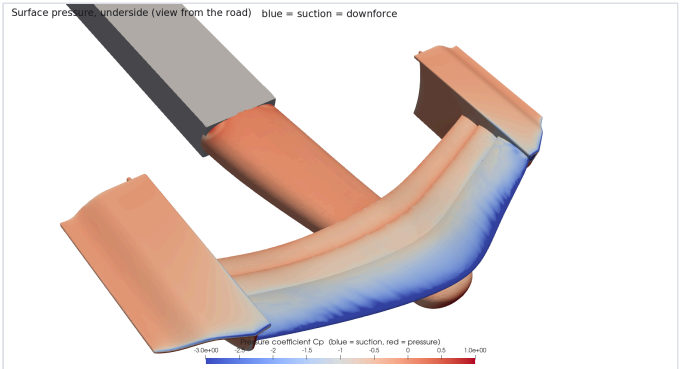


Stopped at iteration 650 (iteration cap) after 650 iterations (37 min wall time). Forces oscillate around a steady mean, typical of steady RANS on a coarse multi-element mesh; reported values are the mean of the last 162 iterations. Flow-field images use the iteration-600 solution.

WHERE THE DOWNFORCE IS MADE

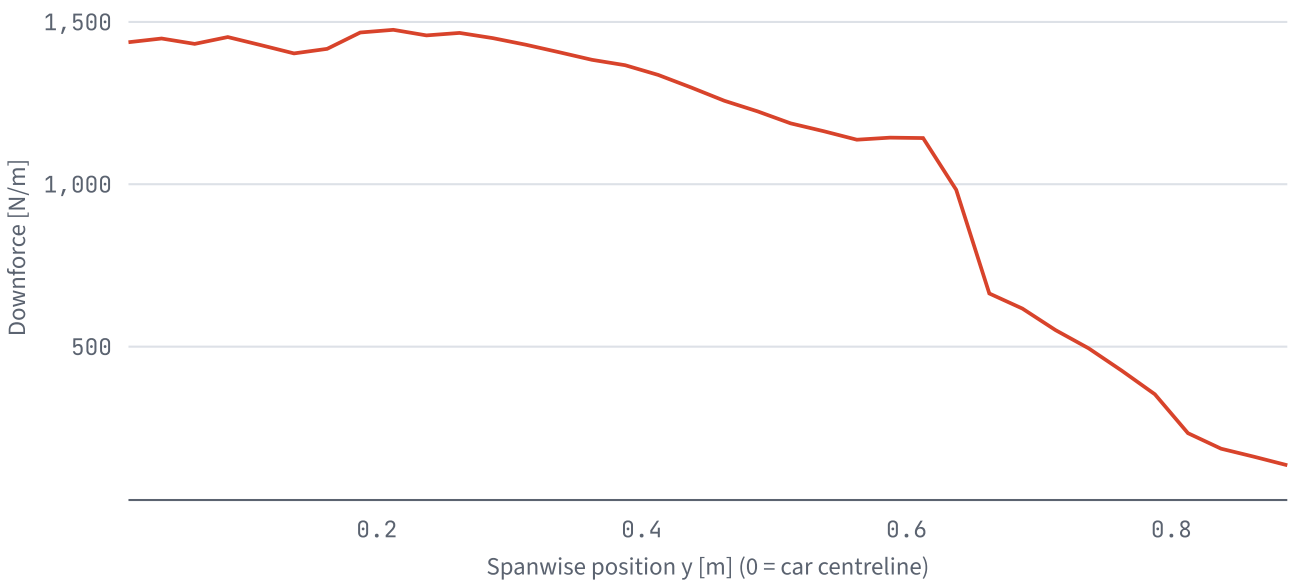


Top side pressure. Red areas are pushed down by the oncoming air; the strongest red sits on the leading edges, where the air hits the wing head-on. Together with the underside suction, this pressure difference creates downforce.



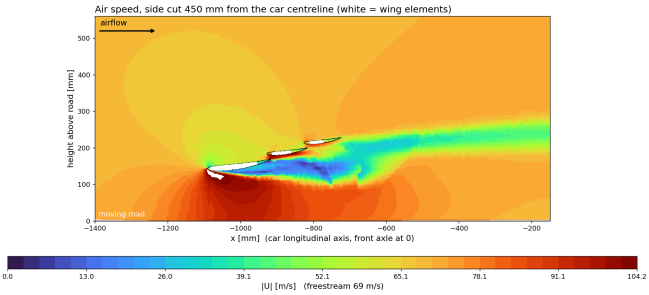
Underside suction, seen from the road. Deep blue marks the strongest suction, under the mainplane and the flap leading edges. Blue regions do most of the work: the closer the wing runs to the road, the stronger this suction (ground effect).

SPANWISE DOWNFORCE DISTRIBUTION

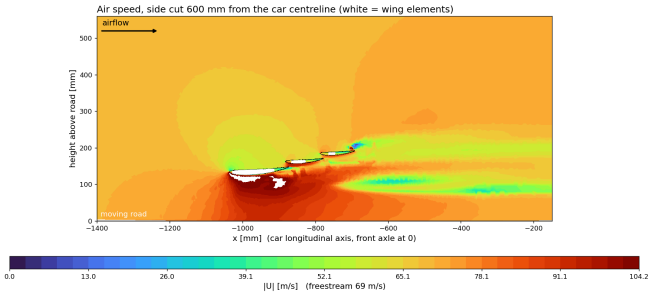


Pressure downforce per metre of span on one side of the car, integrated in 25 mm strips. The y = 0–0.17 m band includes the nose and the neutral centre section of the mainplane.

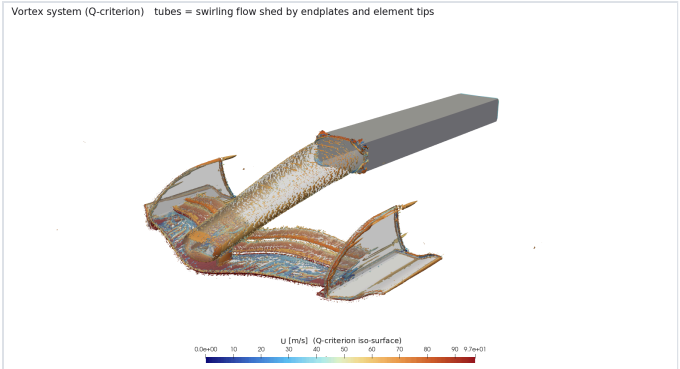
WHAT THE AIR IS DOING



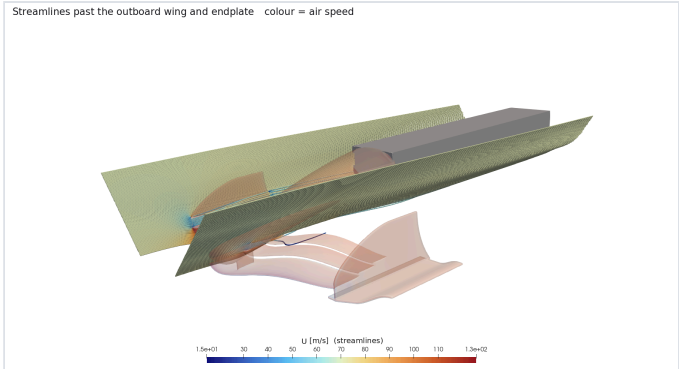
Side cut through the mid-wing. Red is fast air, dark blue slow air. Air speeds up under each element (suction) and jets through the slots between elements, which keeps the flow attached on the flaps. The slow band behind the wing is its wake.



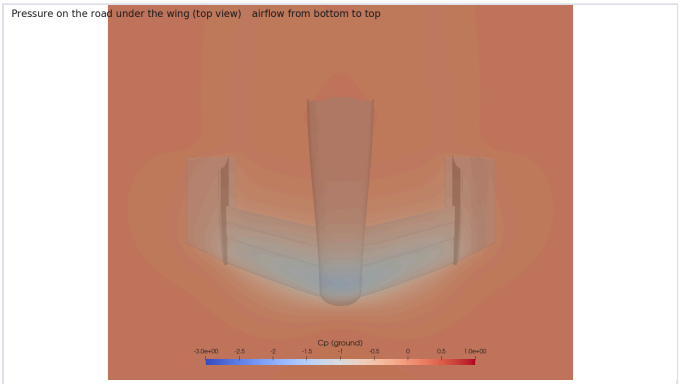
Side cut through the outboard wing, 600 mm from the centreline. Compare with the 450 mm cut: the outboard elements sit higher and further from the road, so the under-wing acceleration is weaker.



Vortex system. Each tube is a vortex (swirling air) shed by an edge: endplate top and bottom, footplate and flap tips. On a full car these vortices steer air around the front tyres and into the floor, so their position matters as much as the downforce number.



Air paths past the endplate. Lines show where air travels. Watch them wind into a spiral behind the endplate (the endplate vortex) and get pushed outboard: this is the outwash F1 designers use to move the front-tyre wake away from the car.



Pressure on the road under the wing. The blue footprint shows the wing sucking on the road surface. Its strength shows how much the wing relies on ground effect, and therefore how sensitive it will be to ride height.

MESH AND SOLUTION QUALITY

METRIC	VALUE
Cells (half model)	2,824,512
Wing surface cell size	3.1 mm (level 6), 1.6 mm on curved edges and features (level 7); 12.5 mm box around the wing, 25 mm wake box
Prism layers	3 on the wing, ER 1.3 (coverage 92 % of faces, 2.8 layers avg)
Max non-orthogonality / skewness	64.9° / 3.46
Wing y^+ (average)	52: wall-function range. The maximum (430) sits in the few junction cells held by the limiter
Meshing / solve wall time	4 min / 37 min on 16 cores

LIMITATIONS AND NEXT STEPS

- Isolated wing with a nose stub. No front wheels, suspension or car body, so wheel–wake and outwash interaction (a primary F1 front wing design driver) is absent.
- Concept mesh with wall functions ($y^+ \approx 52$). Thin trailing edges and the slot gaps are resolved with only a few cells, so separation onset and peak suction are approximate. For sign-off, refine to level 8 on the flap trailing edges and use 10+ layers at $y^+ \approx 1$ with low-Re SST.
- Steady RANS on a single ride height and no yaw or roll. Front wing performance is ride-height sensitive; a ride-height sweep (e.g. 60–110 mm lowest point) is the most valuable next study.
- Half model with symmetry: asymmetric vortex behaviour and yaw cannot be captured.

> Use this result to rank geometry changes on the same mesh settings. Treat the absolute downforce as indicative until a refined-mesh or full-car simulation confirms it.